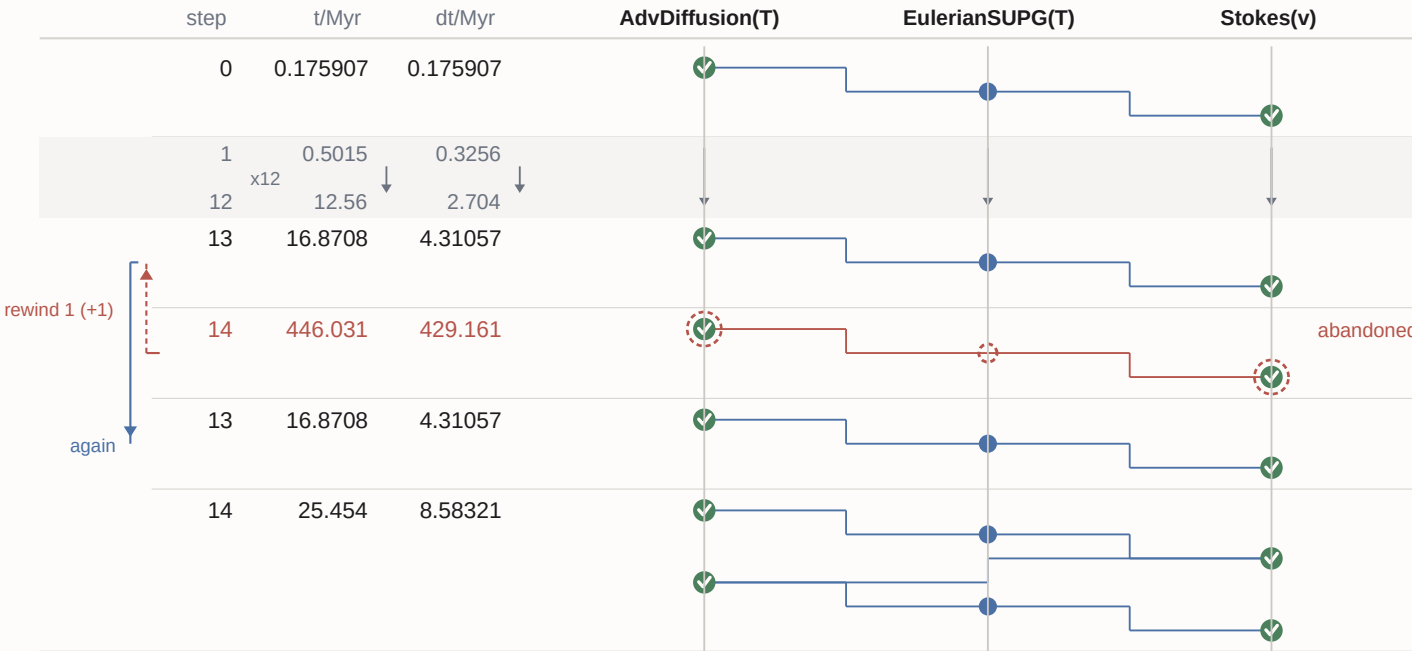


Annulus convection - transcript

started 2026-09-15T14:32:34+10:00 - no terminator: still running, or interrupted - 3 parts



- ran; a step reads down, the path joining what ran in the order it ran
- solved, and converged
- converged, but a fieldsplit block hit its iteration cap - that block did not solve
- did not converge; the reason is in the record
- ran in a step that was then abandoned
- did nothing in this step
- the steps between did exactly this, unchanged; first and last values are shown where they differ

Key - the parts, and what is in their residuals

AdvDiffusion(T) recorded at step 0

$\int F_0 \phi + F_1 \cdot \nabla \phi = 0$ - with

Strong residual of the time scheme: time derivative, advection and source.

$f_0(\phi) = a_{0,18}^{AM}(T_{,0}(\mathbf{x})v_0(\mathbf{x}) + T_{,1}(\mathbf{x})v_1(\mathbf{x})) + a_{1,18}^{AM}(v_0(\mathbf{x})T^*_{,0}(\mathbf{x}) + v_1(\mathbf{x})T^*_{,1}(\mathbf{x})) + \frac{T(\mathbf{x}) - T^*(\mathbf{x})}{\Delta t_{18}}$

$\Delta t_{18} = 1.147 \times 10^{-6}$ - timestep of this history

$a_{0,18}^{AM} = \frac{1}{2}$ - Adams-Moulton coefficient 0 of history 18

$a_{1,18}^{AM} = \frac{1}{2}$ - Adams-Moulton coefficient 1 of history 18

Diffusive flux of the time scheme plus the SUPG flux tau R u.

$$\mathbf{F}_1(\phi) = \begin{bmatrix} \kappa a_{0,18}^{AM} T_{,0}(\mathbf{x}) + \kappa a_{1,18}^{AM} T^*_{,0}(\mathbf{x}) + \frac{w_{18}^{SUPG}(v_0^2(\mathbf{x}) + v_1^2(\mathbf{x})) \left(a_{0,18}^{AM}(T_{,0}(\mathbf{x})v_0(\mathbf{x}) + T_{,1}(\mathbf{x})v_1(\mathbf{x})) + a_{1,18}^{AM}(v_0(\mathbf{x})T^*_{,0}(\mathbf{x}) + v_1(\mathbf{x})T^*_{,1}(\mathbf{x})) + \frac{T(\mathbf{x}) - T^*(\mathbf{x})}{\Delta t_{18}} \right) h_{cell}^2(\mathbf{x}) v_0(\mathbf{x})}{(64\kappa^2 + \varepsilon + (v_0^2(\mathbf{x}) + v_1^2(\mathbf{x}))h_{cell}^2(\mathbf{x})) \sqrt{\frac{\kappa^2(C_{t,18}^T)^2}{h_{cell}^4(\mathbf{x})} + \frac{(C_{t,18}^T)^2}{\Delta t_{18}^2} + \frac{(C_{u,18}^T)^2(v_0^2(\mathbf{x}) + v_1^2(\mathbf{x}))}{h_{cell}^2(\mathbf{x})}} + \varepsilon} \\ \kappa a_{0,18}^{AM} T_{,1}(\mathbf{x}) + \kappa a_{1,18}^{AM} T^*_{,1}(\mathbf{x}) + \frac{w_{18}^{SUPG}(v_0^2(\mathbf{x}) + v_1^2(\mathbf{x})) \left(a_{0,18}^{AM}(T_{,0}(\mathbf{x})v_0(\mathbf{x}) + T_{,1}(\mathbf{x})v_1(\mathbf{x})) + a_{1,18}^{AM}(v_0(\mathbf{x})T^*_{,0}(\mathbf{x}) + v_1(\mathbf{x})T^*_{,1}(\mathbf{x})) + \frac{T(\mathbf{x}) - T^*(\mathbf{x})}{\Delta t_{18}} \right) h_{cell}^2(\mathbf{x}) v_1(\mathbf{x})}{(64\kappa^2 + \varepsilon + (v_0^2(\mathbf{x}) + v_1^2(\mathbf{x}))h_{cell}^2(\mathbf{x})) \sqrt{\frac{\kappa^2(C_{t,18}^T)^2}{h_{cell}^4(\mathbf{x})} + \frac{(C_{t,18}^T)^2}{\Delta t_{18}^2} + \frac{(C_{u,18}^T)^2(v_0^2(\mathbf{x}) + v_1^2(\mathbf{x}))}{h_{cell}^2(\mathbf{x})}} + \varepsilon} \end{bmatrix}$$

$C_{k,18}^T = 4$ - tau diffusive weight

$C_{t,18}^T = 2$ - tau transient weight

$C_{u,18}^T = 2$ - tau advective weight

$\kappa = 10^{-6}$ m²/s - Diffusivity

$\varepsilon = 10^{-18}$ - vanishingly small value

$w_{18}^{SUPG} = 1$ - SUPG term weight (0 = Galerkin)

essential on Lower: 1

essential on Upper: 0

Stokes(v) recorded at step 0

$\int F_0 \phi + F_1 \cdot \nabla \phi = 0$ - with

Velocity equation body force term (pointwise).

$$\mathbf{f}_0(\mathbf{u}) = \begin{bmatrix} \frac{x \rho_0 \alpha g T(\mathbf{x})}{\sqrt{x^2 + y^2}} \\ \frac{y \rho_0 \alpha g T(\mathbf{x})}{\sqrt{x^2 + y^2}} \end{bmatrix}$$

$\rho_0 \alpha g = 0.9712$ kg/K/m²/s² - buoyancy coefficient: reference density x thermal expansivity x gravity

Velocity equation flux/stress term (pointwise).

$$\mathbf{F}_1(\mathbf{u}) = \begin{bmatrix} \eta \lambda (v_{0,0}(\mathbf{x}) + v_{1,1}(\mathbf{x})) + 2\eta v_{0,0}(\mathbf{x}) - p(\mathbf{x}) & 2\eta \left(\frac{v_{0,1}(\mathbf{x})}{2} + \frac{v_{1,0}(\mathbf{x})}{2} \right) \\ 2\eta \left(\frac{v_{0,1}(\mathbf{x})}{2} + \frac{v_{1,0}(\mathbf{x})}{2} \right) & \eta \lambda (v_{0,0}(\mathbf{x}) + v_{1,1}(\mathbf{x})) + 2\eta v_{1,1}(\mathbf{x}) - p(\mathbf{x}) \end{bmatrix}$$

$\eta = 10^{22}$ Pa · s - Shear viscosity

$\lambda = 0$ - Numerical Penalty

Pressure equation constraint term (continuity).

$\mathbf{h}_0(\mathbf{p}) = v_{0,0}(\mathbf{x}) + v_{1,1}(\mathbf{x})$

rotated free-slip on Upper: u · n = 0

rotated free-slip on Lower: u · n = 0